Parameter-Efficient Fine-Tuning for Czech-to-English Machine Translation

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Abstract

Adapting large multilingual machine translation models to specific language pairs, tasks or domains often requires considerable computational resources. This project investigates parameter-efficient fine-tuning (PEFT) as a practical alternative to full fine-tuning for Czech-to-English machine translation. We compare LoRA, IA3, and prefix tuning using `facebook/nllb-200-distilled-600M` and a subset of WMT18 Czech-English, under a low resource environment. All methods are evaluated against a zero-shot baseline and full fine-tuning. In addition to automatic evaluation with BLEU, chrF, and COMET, we examine a range of qualitative translation errors.

The results show that PEFT methods can closely approach full fine-tuning quality while updating less than 0.2% of the model parameters. LoRA provides the strongest overall PEFT performance, while IA3 and prefix tuning offer competitive quality with even fewer trainable parameters. The qualitative analysis further suggests that many remaining errors reflect broader Czech-to-English translation challenges rather than failures of a specific adaptation method. Overall, the study supports PEFT as an effective and resource-conscious approach for adapting multilingual translation models in data and compute-constrained settings.

1 Introduction

Large multilingual machine translation models such as NLLB-200 [3] have made it possible to translate across many language pairs with supervised or even zero-shot performance. However, adapting these models to a specific language pair, domain, or task remains costly when all parameters are updated. Full fine-tuning requires substantial GPU memory, training time, and storage, and it can also increase the risk of overfitting.

Parameter-efficient fine-tuning offers a lighter alternative by keeping most pretrained model weights frozen and training only a small set of additional or modified parameters. This makes PEFT especially relevant and interesting to explore. In this project, we evaluate three PEFT methods: LoRA, IA3, and prefix tuning.

Czech-to-English translation provides a strong test case for parameter-efficient adaptation. Czech is morphologically rich and structurally different from English, with frequent inflection, flexible word order, ellipsis, and context-dependent meaning. These properties make Czech-English translation a useful setting for evaluating whether lightweight adaptation methods can handle linguistic complexity without updating the full model.

Our contributions are threefold. First, we provide a controlled comparison of three PEFT methods using the same base model, dataset split, and evaluation setup. Second, we compare translation quality, parameter efficiency, and implementation trade-offs. Third, we provide a qualitative error analysis focused on Czech-specific translation challenges.

2 Background and Related Work

Recent work has shown that parameter-efficient fine-tuning methods can match or approach full fine-tuning performance across a variety of NLP tasks [5]. LoRA [6] has become one of the most widely adopted PEFT methods due to its strong performance and implementation simplicity. IA3 [8] further reduces trainable parameters by learning activation-scaling vectors rather than low-rank weight updates. Prefix tuning [7] adapts pretrained models by learning virtual prompt representations while keeping model weights frozen. Prior work has demonstrated the effectiveness of PEFT methods for multilingual generation and low-resource

adaptation, and recent work has also proposed unified views of parameter-efficient transfer learning [5]. However, comparative evaluations for Czech-to-English translation with NLLB remain limited.

3 Experimental Setup and Methods

We use a subset of the WMT18 Czech-English dataset [2] from Hugging Face (charlie0831/wmt18-cs-en-preprocessed). In this dataset, each example contains a Czech source sentence and an English reference translation. We remove sentence pairs with empty, non-string, or very short fields, while preserving Czech diacritics and UTF-8 in preprocessing. To simulate a data-constrained adaptation setting, we use 48,000 sentence pairs for training. We use 1,000 examples for validation and 1,000 examples for final testing, with random seed 42. The same train, validation, and test splits are used for all systems. All systems use the `facebook/nllb-200-distilled-600M` base model [3] and the corresponding NLLB tokenizer. Maximum sequence length is set to 512 tokens for both source and target. At inference time, all systems use beam search with beam size 4 and a maximum of 256 generated tokens. Translation quality is evaluated on the same 1,000-sentence test set using BLEU [9], chrF [10], and COMET [12]. BLEU and chrF are computed with sacreBLEU [11], while COMET is computed with `unbabel-comet`.

We compare five systems. The zero-shot baseline uses the pretrained NLLB model without adaptation. Full fine-tuning updates all model parameters for 3 epochs with batch size 128, learning rate 5×10^{-5}, 500 warmup steps, weight decay 0.01, fp16 mixed precision, and gradient checkpointing. LoRA trains low-rank updates in selected attention projection layers while freezing the base model; the best configuration uses rank 8, alpha 32, query/value target modules, learning rate 3×10^{-4}, and 1 epoch. IA3 trains activation-scaling vectors in attention and feed-forward modules; the best configuration uses key, value, and `wi_1` modules, learning rate 1×10^{-2}, and 3 epochs. Prefix tuning trains virtual prefix tokens while keeping the base model frozen; the best efficiency setting uses 20 virtual tokens, learning rate 5×10^{-3}, a linear scheduler, and 3 epochs.

Because experiments are based on single runs and a limited hyperparameter search, small differences between systems should be interpreted cautiously.

4 Results

4.1 Main Results

Table 1: Main test-set results for Czech-to-English translation.

System	Trainable Parameters	BLEU	chrF	COMET
NLLB zero-shot	0	23.04	50.11	0.741
Full fine-tuning	600M (100%)	31.15	57.47	0.833
LoRA	1.18M (0.19%)	31.70	57.82	0.831
IA3	560K (0.09%)	30.26	56.83	0.828
Prefix tuning	491K (0.08%)	30.55	56.98	0.828

All fine-tuned systems substantially outperform the zero-shot baseline. LoRA obtains the highest observed BLEU and chrF scores, slightly exceeding full fine-tuning on lexical metrics while updating only 0.19% of the model parameters. Full fine-tuning obtains the highest observed COMET score, narrowly above LoRA. Prefix tuning and IA3 trail LoRA by roughly 1.0 to 1.5 BLEU points but remain very close to full fine-tuning.

The differences among the fine-tuned systems are relatively small, especially for COMET. Therefore, these results should be interpreted as evidence that PEFT methods can closely match full fine-tuning in this setting, rather than as a definitive ranking of methods.

5 Parameter Comparison Between Methods

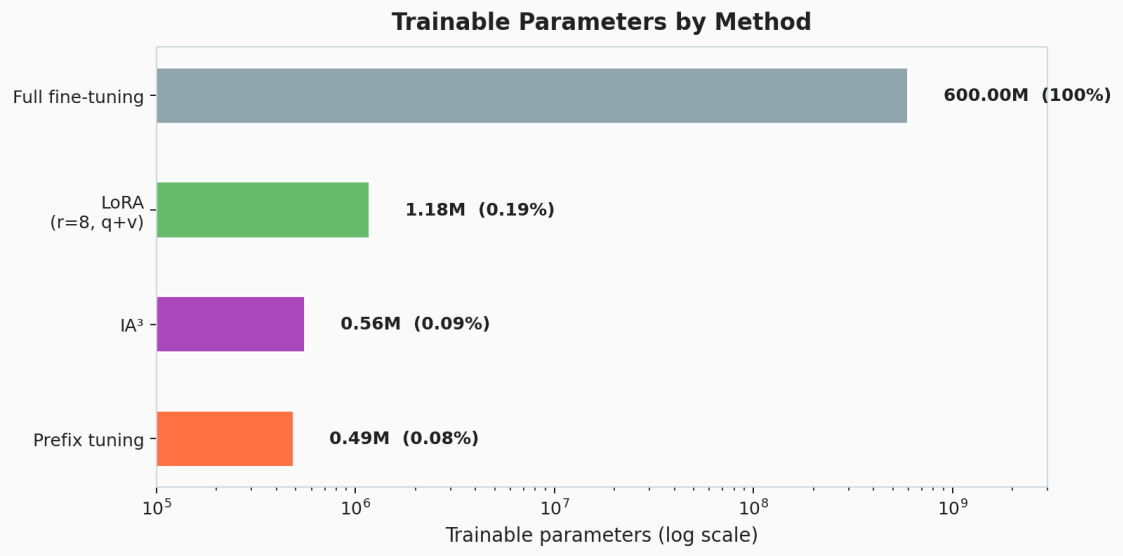


Figure 1: Parameter comparison between full fine-tuning and parameter-efficient fine-tuning methods.

5.1 LoRA Results

The best LoRA configuration uses rank 8, alpha 32, learning rate 3×10^{-4}, query/value target modules, dropout 0.1, and 1 epoch. This setting reaches 31.70 BLEU, 57.82 chrF, and 0.831 COMET. A table with full runs can be found in the Appendix B. The LoRA ablation suggests that increasing training duration does not necessarily improve lexical metrics. For example, the same rank-8, alpha-32, query/value configuration trained for 3 epochs with learning rate 3×10^{-4} reaches 31.27 BLEU, below the 1-epoch result, although COMET remains slightly higher in some longer runs. Additional training may improve semantic adequacy or fluency while moving outputs away from the reference. Increasing the rank to 16 does not produce a clear gain, and expanding the target modules from query/value projections to all attention projections also does not improve performance. Overall, the results indicate that even a standard LoRA configuration is sufficient to get good results.

5.2 IA3 Results

For all IA3 experiments, we train for 3 epochs with the same data split and evaluation setup. We vary the learning rate across three values: 5×10^{-4}, 3×10^{-3}, and 1×10^{-2}.

IA3 is sensitive to learning rate. At 5×10^{-4} and 3×10^{-3}, the model underfits, reaching 27.61 and 26.24 BLEU respectively. At 1×10^{-2}, IA3 reaches 30.26 BLEU, 56.83 chrF, and 0.828 COMET, getting closer to the stronger PEFT methods. This makes IA3 attractive from a parameter-efficiency perspective, but less stable without learning-rate tuning.

5.3 Prefix Tuning Results

Prefix tuning achieves competitive performance while training fewer than 0.2% of the model parameters across all tested configurations. The best efficiency setting uses 20 virtual tokens, 3 epochs, learning rate 5×10^{-3}, and a linear scheduler, reaching 30.55 BLEU, 56.98 chrF, and 0.828 COMET with only 491K trainable parameters.

Increasing the prefix length to 50 virtual tokens and training for 5 epochs improves BLEU only slightly, from 30.55 to 30.65, and COMET from 0.828 to 0.830. Given the additional parameters and training time, the smaller 20-token configuration offers the better quality-efficiency trade-off. A lower learning rate with a

cosine scheduler performs nearly identically to the main 20-token setting, suggesting that prefix tuning is relatively robust across the tested hyperparameters.

5.4 Cross-Metric Patterns and Outliers

Metric disagreement between BLEU and COMET was observed across several runs. While BLEU penalizes surface-level lexical and word-order variation, COMET is more sensitive to semantic adequacy and fluency. Similar disagreements between lexical and neural MT evaluation metrics have been discussed in prior work [4]. In the main results, LoRA achieves the highest BLEU and chrF scores, while full fine-tuning achieves the highest COMET score. However, the absolute differences are small: the COMET range is narrow across fine-tuned systems, and the BLEU differences among the strongest systems are also modest. These gaps should therefore not be over-read without repeated runs or statistical testing.

6 Error Analysis

This section presents a qualitative analysis of translation errors across all four systems: Prefix Tuning, IA3, LoRA, and Full Fine-tuning. We manually inspect low-scoring examples using chrF as a guide, grouping errors according to recurring linguistic and semantic patterns. Full examples are available in the Appendix C, which also includes a cross-system comparison conducted as an additional analysis.

6.0.1 Linguistic Error Categories

The following error types reflect inherent challenges of Czech-to-English machine translation. While not all errors appear in every system, the categories below represent patterns observed across multiple methods.

Proper nouns and named entities are a recurring source of error. Czech place names and proper nouns are often morphologically inflected, which makes them difficult to recognize. Prefix tuning and IA3 translated *Kolínsko* (a Czech region) as *Cologne*, confusing a Czech toponym with a visually similar German city name (see Appendix C, Table 3). Similarly, in the sentence *„Spojení Dědka se STAN?"* (*„Is there a link between Dědek and STAN?"*), Prefix tuning translated the politician's surname *Dědek* (nominativ form of genitiv *Dědka*) as *Grandfather* and the political party abbreviation *STAN* as a common noun *Stan*, losing both the proper name and the abbreviation.

Czech allows heavy **ellipsis**; subjects, verbs, and contextual information are frequently omitted when implied by context. All systems translate what is present but cannot recover the missing meaning. The sentence *„Ten s dvaceti chudý"* (*„The one with twenty thousand [crowns] is poor"*) omits both the verb and the currency unit; none of the systems recovered the full intended meaning.

Czech **idioms** are often translated literally, losing the intended meaning. The idiom *„přilévat olej do ohně"*, meaning *„to add fuel to the fire"*, was left untranslated by full fine-tuning, which copied the Czech source verbatim.

Several errors arise from Czech words that visually resemble English words but carry different meanings, a phenomenon known as **false friends**. *„Polidštit"* (meaning *„to humanize"*), was translated as *„to polish"* by LoRA, producing a fluent but incorrect output. *„Fandí"* (meaning *„to cheer for"* or *„to support"*) was confused with *„fanatic"*.

Several sentences require knowledge of the surrounding text to be translated correctly, a limitation referred to as **missing discourse context**. The sentence *„Jeho podmořská hrobka zachovala tělo v dokonalém stavu"* describes a dinosaur fossil, but without context all systems defaulted to a human burial place, producing translations that read as fluent but are factually wrong.

6.0.2 Key Findings

The error analysis reveals several important patterns. Shared error types, including proper nouns, ellipsis, idioms, false friends, and missing discourse context, affect multiple systems, reflecting fundamental challenges of Czech-to-English MT rather than weaknesses of any specific method.

Full fine-tuning exhibits the most severe method-specific errors: **Czech leakage** and **hallucination**. These errors were not observed in the inspected PEFT examples, suggesting that PEFT approaches may

be more stable and predictable even when they fail. LoRA shows no distinct error type but has the lowest worst-case chrF range across all systems, indicating higher instability on difficult sentences. High-confidence wrong translations, where fluent output carries incorrect meaning, are the most dangerous error type from a practical standpoint because they may appear correct to users without knowledge of the source language.

Finally, some low chrF scores reflect valid **paraphrases** penalized for differing from the reference wording rather than genuine translation errors, which is a known limitation of n-gram-based metrics. A summary of method-specific error severity is provided in Appendix E, Table 4 and 5.

7 Discussion

The main result is that PEFT methods can closely approach full fine-tuning quality while updating only a small fraction of the model. One possible explanation is that NLLB already contains strong Czech-English representations from pretraining, so adaptation mainly needs to steer the model toward the target dataset rather than learn translation from scratch. Freezing most weights may also act as regularization, preventing the model from overwriting useful multilingual knowledge. The differences between PEFT methods reflect their adaptation mechanisms. LoRA performs best in this study, likely because it can directly change linear transformations through low-rank updates. IA3 is cheaper but less expressive because it only rescales existing activations. Prefix tuning is simple and efficient, but appears to plateau below LoRA because it influences the model indirectly through attention prefixes.

The metric disagreement matters practically. If exact reference overlap is important, LoRA appears strongest. If semantic adequacy and fluency are prioritized, full fine-tuning remains slightly ahead. Since the differences are small, reporting only one metric would give an incomplete picture. Prior work has shown that BLEU and neural metrics can systematically disagree and that neither is unconditionally better [4]. For Czech specifically, this tension is further complicated by inflection.

LoRA achieves the highest lexical scores and is the most stable across configurations, but it is the slowest of the three PEFT methods to train and may require adapter merging or adapter management at inference time. IA3 uses very few parameters and trains quickly, making it well suited to settings where many adapters need to be stored or served from a shared model; its main drawback is sensitivity to learning rate. Prefix tuning is simple and robust, but it plateaus below LoRA regardless of prefix length. This is consistent with the view that fine-tuning can often operate in a low-dimensional task-specific subspace [1].

8 Conclusion

This project evaluated whether parameter-efficient fine-tuning can serve as a practical alternative to full fine-tuning for Czech-to-English machine translation. Using `facebook/nllb-200-distilled-600M` on a data-constrained subset of WMT18, we found that LoRA, IA3, and prefix tuning all substantially improve over the zero-shot baseline while updating less than 0.2% of the model parameters. LoRA achieved the strongest PEFT results on BLEU and chrF, while full fine-tuning remained slightly ahead on COMET. However, the differences among the adapted systems were small, suggesting that PEFT methods can offer a strong quality-efficiency trade-off in this setting.

The qualitative analysis shows that many remaining errors are linked to Czech-specific translation challenges, including inflection, named entities, ellipsis, idioms, false friends, and missing discourse context. These patterns suggest that the main limitation is not only the adaptation method, but also the difficulty of recovering meaning from linguistically complex and context-dependent inputs.

Overall, our results support PEFT as a resource-conscious approach for adapting multilingual machine translation models when compute, memory, or storage are limited. Future work should evaluate these methods with repeated runs, broader hyperparameter tuning, larger training subsets, and additional Czech-English domains to determine whether the observed trends remain stable at larger scale.

Limitations

This study is limited by compute, dataset size, and the scope of hyperparameter search. We evaluate on a 48,000-sentence subset of WMT18 rather than the full dataset, and each PEFT method is tested with only a small number of configurations, mostly using standard PEFT implementations. Results are also based on single runs, so small differences between systems should not be over-interpreted without statistical testing.

Future work could focus on the quality of fine-tuning data, as the WMT18 dataset contains noisy examples including offensive content and morphological errors in proper nouns, which may introduce undesirable outputs.

Ethics Statement

This project uses a public machine translation dataset and a publicly available pretrained model. No private or personally collected data was used. However, machine translation systems can produce incorrect or misleading translations, especially for ambiguous, idiomatic, or context-dependent sentences. Outputs should not be treated as reliable in high-stakes settings without human review.

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References

[1] Armen Aghajanyan, Luke Zettlemoyer, and Sonal Gupta. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Proceedings of ACL-IJCNLP 2021*, pages 7319–7328, 2021.

[2] Ondřej Bojar, Christian Federmann, Mark Fishel, Yvette Graham, Barry Haddow, Philipp Koehn, and Christof Monz. Findings of the 2018 conference on machine translation (wmt18). In *Proceedings of the Third Conference on Machine Translation*, pages 272–303, 2018.

[3] Marta R. Costa-jussà et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022.

[4] Taisiya Glushkova, Chrysoula Zerva, Ricardo Rei, and André F. T. Martins. Revisiting mt evaluation: There is no simple fix. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1016–1031, 2023.

[5] Junxian He, Chunting Zhou, Xuezhe Ma, Taylor Berg-Kirkpatrick, and Graham Neubig. Towards a unified view of parameter-efficient transfer learning. In *Proceedings of ICLR 2022*, 2022.

[6] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *Proceedings of ICLR 2022*, 2022.

[7] Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. In *Proceedings of ACL-IJCNLP 2021*, pages 4582–4597, 2021.

[8] Haokun Liu, Derek Tam, Mohammed Muqeeth, Jay Mohta, Tenghao Huang, Mohit Bansal, and Colin Raffel. Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning. In *Proceedings of NeurIPS 2022*, 2022.

[9] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: A method for automatic evaluation of machine translation. In *Proceedings of ACL 2002*, pages 311–318, 2002.

[10] Maja Popović. chrf: Character n-gram f-score for automatic mt evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, 2015.

[11] Matt Post. A call for clarity in reporting bleu scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191, 2018.

[12] Ricardo Rei, Craig Stewart, Ana C. Farinha, and Alon Lavie. Comet: A neural framework for mt evaluation. In *Proceedings of EMNLP 2020*, pages 2685–2702, 2020.

Appendix

This appendix reports supplementary experimental details, full ablation results, and extended qualitative error analysis. These materials support the main findings but are separated from the main text for readability.

A Contributor Roles

Table 2: Division of work across contributors.

Contributor	**Main responsibilities**
Andreia Alexa	LoRA fine-tuning experiment including dependencies; methodology proposal and direction, coordinated report writing
Jie Huang	Zero-shot and full fine-tuning baselines; shared evaluation script; standardized project dependencies.
Yu Chien Lin	IA3 fine-tuning experiment; dataset extraction and preparation.
Diana Mikova	Prefix tuning experiment; Error analysis and reporting.
All contributors	Contributed to results evaluation, wrote assigned report sections, reviewed the report, and prepared presentation slides.

B LoRA Ablation Results

All LoRA runs are sorted by BLEU. The checkpoint run evaluates an intermediate checkpoint at approximately 1,000 steps, or roughly 0.3 epochs, rather than a final model. LoRa works on the assumption that models contain a low-dimensional structure. By leveraging low-rank (smaller) matrices, LoRA adapts these models to specific task or dataset.

The rank-4 run used only 589K parameters and still reached 31.44 BLEU, close to the best rank-8 result of 31.70 BLEU. This shows one of LoRA's main strengths: you can reduce trainable parameters a lot and still get strong performance.

Rank controls the capacity of the LoRA adapter by setting the size of the low-rank matrices, while alpha controls the strength of the adapter update through the scaling factor alpha/r. In the best LoRA run, r = 8 and alpha = 32, so the adapter used 1.18M trainable parameters and its learned correction was scaled by $32/8 = 4$ before being added to the frozen NLLB projection.. The matrix size is the same in all three cases, but the correction is added with different strength.

Table 3: Full LoRA ablation results sorted by BLEU.

Experiment	r	α	LR	Targets	Dropout	Ep.	Eval Loss	BLEU	chrF	COMET
r8-a32-lr3e4-qv-1epoch	8	32	3e-4	q,v	0.1	1	1.2608	31.70	57.82	0.831
r8-a32-lr1e4-qv-3epochs	8	32	1e-4	q,v	0.1	3	1.2621	31.54	57.71	0.832
r4-a16-lr5e4-qv-1epoch	4	16	5e-4	q,v	0.1	1	1.2593	31.44	57.75	–
r8-a32-lr3e4-qv-3epochs	8	32	3e-4	q,v	0.1	3	1.2553	31.27	57.53	0.832
r4-a16-lr5e4-qv-3epochs	4	16	5e-4	q,v	0.1	3	1.2558	31.26	57.60	–
r16-a32-lr5e4-qv-3epochs	16	32	5e-4	q,v	0.1	3	1.2573	31.26	57.58	0.832
r4-a16-lr5e4-qv-ckpt	4	16	5e-4	q,v	0.1	0.3	1.2795	31.29	57.62	0.830
r8-a32-lr5e4-allattn	8	32	5e-4	q,k,v,o	0.1	1	1.2552	31.20	57.63	0.831
r8-a32-lr1e4-qv	8	32	1e-4	q,v	0.1	1	1.2795	31.20	57.66	0.831
r8-a32-lr5e4-qv-dropout05	8	32	5e-4	q,v	0.05	1	1.2591	31.19	57.46	0.831
r8-a32-lr5e4-qv-3epochs	8	32	5e-4	q,v	0.1	3	1.2605	31.05	57.46	0.8327

These results show that increasing LoRA rank, training duration, or target-module coverage does not reliably improve BLEU in this setting.

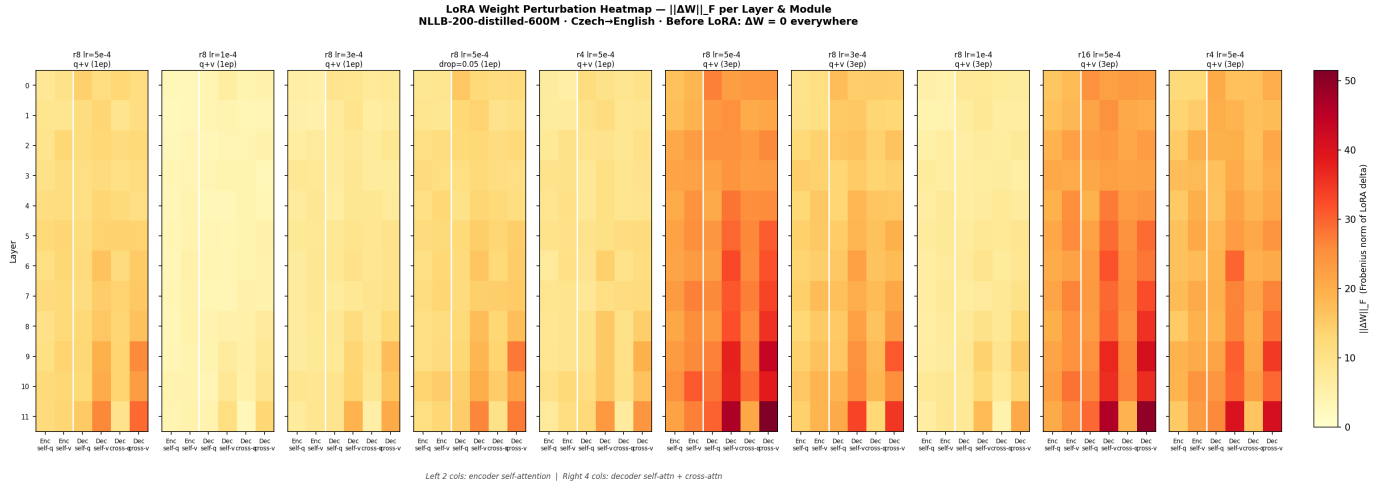


Figure 2: LoRA Weight Perturbation Heatmap per Layer and Module.

C IA3 Hyperparameter Results

Table 4: IA3 learning-rate ablation.

Metric	Exp. 1	Exp. 2	Exp. 3
Epochs	3	3	3
Learning rate	5e-4	3e-3	1e-2
BLEU	27.61	26.24	30.26
chrF	55.30	54.26	56.80
COMET	0.8118	0.8070	0.8280

The IA3 results show strong learning-rate sensitivity, with the highest tested learning rate producing the best translation quality.

D Prefix Tuning Hyperparameter Results

Table 5: Prefix tuning ablation results.

Exp.	Trainable Params	Tokens	Ep.	LR	Scheduler	BLEU	chrF	COMET
1	491K (0.08%)	20	3	5e-3	linear	30.55	56.98	0.828
2	1.2M (0.20%)	50	5	5e-3	linear	30.65	57.00	0.830
3	491K (0.08%)	20	3	1e-3	cosine	30.53	56.99	0.827

E Error Analysis

This appendix presents a detailed linguistic and method-specific error analysis, based on manual inspection of the lowest-scoring outputs from each system using chrF as a ranking criterion.

While most error types are shared across systems, several failure patterns are unique to individual methods. **Prefix tuning** shows one clear hallucination of a proper name: the Czech name *Láďa* was replaced with *'Ark'*, a name that does not appear anywhere in the source. **IA3** shows false friend errors not observed in other systems, as illustrated by the *Fandí všichni* example (see Table 3). **LoRA** does not exhibit a distinct error type, but its worst-case chrF scores are notably lower than those of the other systems, suggesting less stability on difficult sentences. **Full fine-tuning** exhibits two severe error types not observed in any inspected PEFT output: Czech leakage and semantic reversal.

Table 6: Linguistic-specific error examples.

Category	SRC	HYP	REF
Proper noun	Na **Kolínsku** hořelo pole.	A field was burning in **Cologne**.	Fire breaks out in Kolín.
Ellipsis	Ten s dvaceti chudý.	The poor guy with 20.	Those earning 20,000 are poor.
Idiom	Olej do ohně přilévá chudoba.	Olej do ohně přilévá chudoba.	Poverty feeds the discontent.
False friend	Fandí všichni.	They're all fanatics.	Everyone supports them.
Discourse context	Jeho podmořská hrobka zachovala tělo v dokonalém stavu.	His underwater tomb kept the body in perfect condition.	But the dinosaur's undersea burial preserved its armor in exquisite detail.

Table 7: Method-specific error examples.

Category	SRC	HYP	REF
Prefix (hallucination)	O Láďově kulinářském umění si myslí své.	He thinks of Ark's culinary art.	He has his own thoughts on Láďa's cooking.
Full FT (leakage)	Z ramen mu vyčnívaly dva 50 cm ostny.	Z ramen mu vyčnívaly two 50 cm ostny.	It featured two 20-inch-long spikes which protruded from its shoulders.
Full FT (reversal)	To si nepamatuji.	It's not uncommon.	I don't remember that.

Table 8: Method-specific error severity by chrF score range.

Method	**Score Range**	**Main Error Type**	**Severity**
Prefix tuning	7.68–17.49	Pragmatic / named entity	Moderate
IA3	7.68–17.65	False friends / ellipsis	Moderate
LoRA	2.56–3.97	Semantic distortion	High
Full fine-tuning	6.02–12.49	Czech leakage / hallucination	Severe